

Transmission Gear Oil Pressure Sensor Signal Wrong or Missing

Symptoms

- Transmission gear oil pressure value is constant on ED-4 display.

How To Use This Tree

This symptom tree can be used to troubleshoot a transmission gear oil pressure sensor fault. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Possible causes are:

- Transmission gear oil pressure sensor signal is open.
- Transmission gear oil pressure sensor malfunction.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the alarm code.	
	STEP 1A. Check for an active fault code.	Fault Code 1199 or 1214 active or inactive with more than one count logged in the last 25 engine hours?
	STEP 1B. Check for an active alarm code.	Alarm Code 1214 or 1199 active?
STEP 2.	Check the ED-4 display.	
	STEP 2A. Verify sensor data in the ED-4 display.	Measured value match the recorded value?
STEP 3.	Check the sensor wiring harness.	
	STEP 3A. Inspect the wiring harness.	Dirty or damaged pins?
	STEP 3B. Check the circuit response.	Alarm Code 1199 active?

STEPS		SPECIFICATIONS
	STEP 3C. Check for an open circuit in the transmission gear oil pressure signal circuit.	Less than 10 ohms?

STEP 1. Check the alarm code.

STEP 1A. Check for an active fault code.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for active fault code. • Use INSITE™ electronic service tool to read the inactive fault codes.	Fault Code 1199 or 1214 active or inactive with more than one count logged in the last 25 engine hours? YES	Go to the appropriate troubleshooting tree.
	Fault Code 1199 or 1214 active or inactive with more than one count logged in the last 25 engine hours? NO	No repair.

STEP 1B. Check for an active alarm code.

Conditions :

- Turn system enable switch ON.

Action	Specification/Repair	Next Step
Wait 30 seconds, read the ED-4 Display.	Alarm Code 1214 or 1199 active? YES	Go to appropriate alarm code troubleshooting tree in Section TA.
	Alarm Code 1214 or 1199 active? NO	2A

STEP 2. Check the ED-4 display.

STEP 2A. Verify sensor data in the ED-4 display.

Conditions : • Turn system enable switch ON.		
Action	Specification/Repair	Next Step

Check transmission gear oil pressure. - Record the transmission gear oil pressure sensor raw value in the I/O viewer. - Disconnect the ED-4 display from the customer interface box (C.I.B.). Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-023.html) - Disconnect the primary and secondary connectors from the ED-4 display. - Place one lead on transmission gear oil pressure sensor SIGNAL pin 2 on the ED-4 secondary connector. - Place the other lead on the RETURN pin 1 on the ED-4 primary connector. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Measured value match the recorded value? YES	3A
	Measured value match the recorded value? NO Repair : Check ED-4 Display Analog Input channels in the data log and view - I/O viewer section. Refer to Procedure 015-035 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-035.html) Replace ED-4 if analog channel is not working properly. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-023.html)	Repair complete

STEP 3. Check the sensor wiring harness.

STEP 3A. Inspect the wiring harness.

Conditions :

- Turn system enable switch OFF.
- Disconnect the transmission gear oil pressure sensor from the wiring harness.

Action	Specification/Repair	Next Step
Inspect the wiring harness. • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible.	Repair complete
	Dirty or damaged pins? NO	3B

STEP 3B. Check the circuit response.

Conditions :

- Turn system enable switch OFF.
- Disconnect the transmission gear oil pressure sensor from the wiring harness.
- Turn system enable switch ON.

Action	Specification/Repair	Next Step
Inspect the wiring harness. • Place a jumper wire between the transmission Gear oil pressure +5 volt RETURN pin and the transmission gear oil pressure SIGNAL pin at transmission gear oil pressure sensor connector of the harness. Refer to the circuit diagram or wiring diagram for connector pin identification. Check for the appropriate circuit response after 30 seconds. • Wait 30 seconds, read the ED-4 display.	Alarm Code 1199 active? YES Repair : Replace the transmission gear oil pressure sensor. Refer to Procedure 019-641 in Section 19. (/qs3/pubsys2/xml/en/procedures/513/513-019-641.html)	Repair complete
	Alarm Code 1199 active? NO	3C

STEP 3C. Check for an open circuit in the transmission gear oil pressure signal circuit.**Conditions :**

- Turn system enable switch OFF.
- Disconnect the transmission gear oil pressure sensor from the wiring harness.
- Disconnect all wiring harness in series from the transmission gear oil pressure sensor to the C.I.B.

Action	Specification/Repair	Next Step
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Measure the resistance. - Place one lead on the transmission gear oil pressure sensor SIGNAL pin on the wiring harness connector. - Place the other lead on the transmission gear oil pressure sensor SIGNAL pin in the wiring harness alternate connector. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general multimeter usage techniques. Refer to Procedure 019-359 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)	Less than 10 ohms? YES Repair : Replace transmission oil pressure sensor. Refer to Procedure 019-641 in Section 19 (/qs3/pubsys2/xml/en/procedures/513/513-019-641.html).	Repair complete
	Less than 10 ohms? NO Repair : An open in the transmission gear oil pressure sensor signal circuit has been detected. Troubleshoot each harness in series to determine which harness or connector contains the open circuit. Repair or replace the appropriate harness.	Repair complete.